Paleomagnetism of Eocambrian Long Range dykes and Double Mer Formation from Labrador, Canada

G. MURTHY
Department of Earth Sciences, Memorial University of Newfoundland, St. John's, Nfld., Canada A1B 3X5
C. GOWER
Newfoundland Department of Mines and Energy, P.O. Box 8700, St. John's, Nfld., Canada A1B 4J6
AND
M. TUBRETT AND R. PÄTZOLD
Department of Earth Sciences, Memorial University of Newfoundland, St. John's, Nfld., Canada A1B 3X5

Received June 5, 1991
Revision accepted December 24, 1991

A paleomagnetic study was carried out on two post-Grenvillian units in the Grenville Province of southeast Labrador. These are (*i*) a swarm of northeast-trending mafic dykes inferred to be part of the Long Range suite and (*ii*) the Double Mer Formation, which is an undated rift-related red-bed sequence. Three of six mafic dykes investigated, including two having K–Ar (minimum) ages of 514 ± 8 and 553 ± 22 Ma, carry an interpreted primary remanence of declination (D) = 124.8°, inclination (I) = 55.5° (k = 48, α_{95} = 18.0°), with the corresponding paleopole at 10.8°S, 164.3°E. Three dykes, including one previously dated at 615 ± 2 Ma (U–Pb), yielded anomalous results. Nevertheless, because all six dykes show similar field, petrographic, and chemical characteristics, we consider that they belong to a single swarm. The anomalous directions are interpreted as disturbed paleomagnetic signatures, and the 615 Ma age is extrapolated as dating the primary remanence in the undisturbed dykes. The Double Mer Formation yields a mean direction for five sites at D = 110.8°, I = 50.1° (k = 24.6, α_{95} = 15.7°), with a corresponding paleopole at 13.0°S, 178.6°E. The similarity of this direction with that for mafic dykes is the first quantitative indication of its previously suspected Eocambrian age. Combining the Long Range dyke and Double Mer data, the paleolatitude of Eocambrian southeast Labrador is estimated to be 26 ± 5°S.

Une étude paléomagnétique a été effectuée sur deux unités postgrenvilliennes, dans la province de Grenville, du sud-est du Labrador. Ce sont (*i*) un essaim de dykes mafiques, de direction nord-est, faisant partie du cortège de Long Range et (*ii*) la Formation de Double Mer, laquelle représente une séquence de couches rouges, non datée, mais reliée au rift. Trois parmi les six dykes mafiques étudiés, incluant deux qui ont été datés par K–Ar (minimum) de 514 ± 8 et 553 ± 22 Ma, ont fourni une rémanence primaire caractérisée par déclinaison (D) = 124,8°, inclinaison (I) = 55,5° (k = 48, α_{95} = 18,0°), avec un paléopôle correspondant à 10,8°S, 164,3°E. Trois dykes, dont un d'âge connu de 615 ± 2 Ma (U–Pb), ont fourni des résultats anomaux. Cependant, vu que ces six dykes montrent des caractéristiques pétrographiques, chimiques et magnétiques analogues, nous les considérons comme appartenant tous à un même essain. Les directions anomales sont interprétées comme des signatures paléomagnétiques modifiées, et nous concluons que l'âge de 615 Ma date la rémanence primaire acquise dans les dykes non dérangés. La Formation de Double Mer fournit une direction moyenne de cinq sites de D = 110,8°, I = 50,1° (k = 24,6, α_{95} = 15,7°), avec un paléopôle correspondant à 13,0°S, 178,6°E. Cette direction similaire à celles des dykes mafiques est donc la première mesure quantitative de son âge éocambrien, bien que présumé antérieurement. Les données couplées des dykes de Long Range et de la Formation de Double Mer permettent d'estimer la paléolatitude du sud-est du Labrador, pour l'Éocambrien, à 26 ± 5°S.

[Traduit par la rédaction]

Can. J. Earth Sci. **29**, 1224–1234 (1992)

Introduction

This study reports paleomagnetic data for two post-Grenvillian units in the Grenville Province in southeast Labrador. These units are (*i*) a swarm of northeast-trending mafic dykes in the vicinity of Sandwich Bay, which are correlated with the Long Range dykes in northwest Newfoundland; and (*ii*) the Double Mer Formation in the Lake Melville region (Figs. 1 and 2), which is believed to be Eocambrian. The paleomagnetic data reported here are the first available for either group of rocks. The reporting of these data represents, in part, a continuation of our efforts to classify mafic igneous rock from coastal Labrador by the integrated use of field, petrographic, paleomagnetic, and geochemical parameters (cf. Murthy *et al.* 1989*a*, 1989*b*).

In the final section of the article, the contribution of our data to the Eocambrian paleomagnetic record of cratonic North America is evaluated.

Geological setting

Long Range dykes

The term Long Range dykes was first applied by Bostock *et al.* (1983) to a swarm of northeast-trending mafic dykes that intrude Grenvillian rocks of the Long Range Inlier in northwest Newfoundland. For several years, the name has also been informally applied to post-Grenvillian dykes having similar trends in Labrador. Fahrig and West (1986) introduced the term Trunmore Bay Dykes for the dykes in southeast Labrador. We feel, however, that a separate name is not justified and do not use it in this paper.

There is good control on the age of this swarm. A sample from the longest dyke in Labrador (Fig. 2) has yielded an age of 615 ± 2 Ma (U–Pb zircon and baddeleyite; Kamo *et al.* 1989). K–Ar ages of 514 ± 8 Ma (whole rock; Grasty *et al.* 1969) and 553 ± 22 Ma (biotite; Wanless *et al.* 1970) had been previously obtained from two other dykes (Fig. 2). The

Printed in Canada / Imprimé au Canada

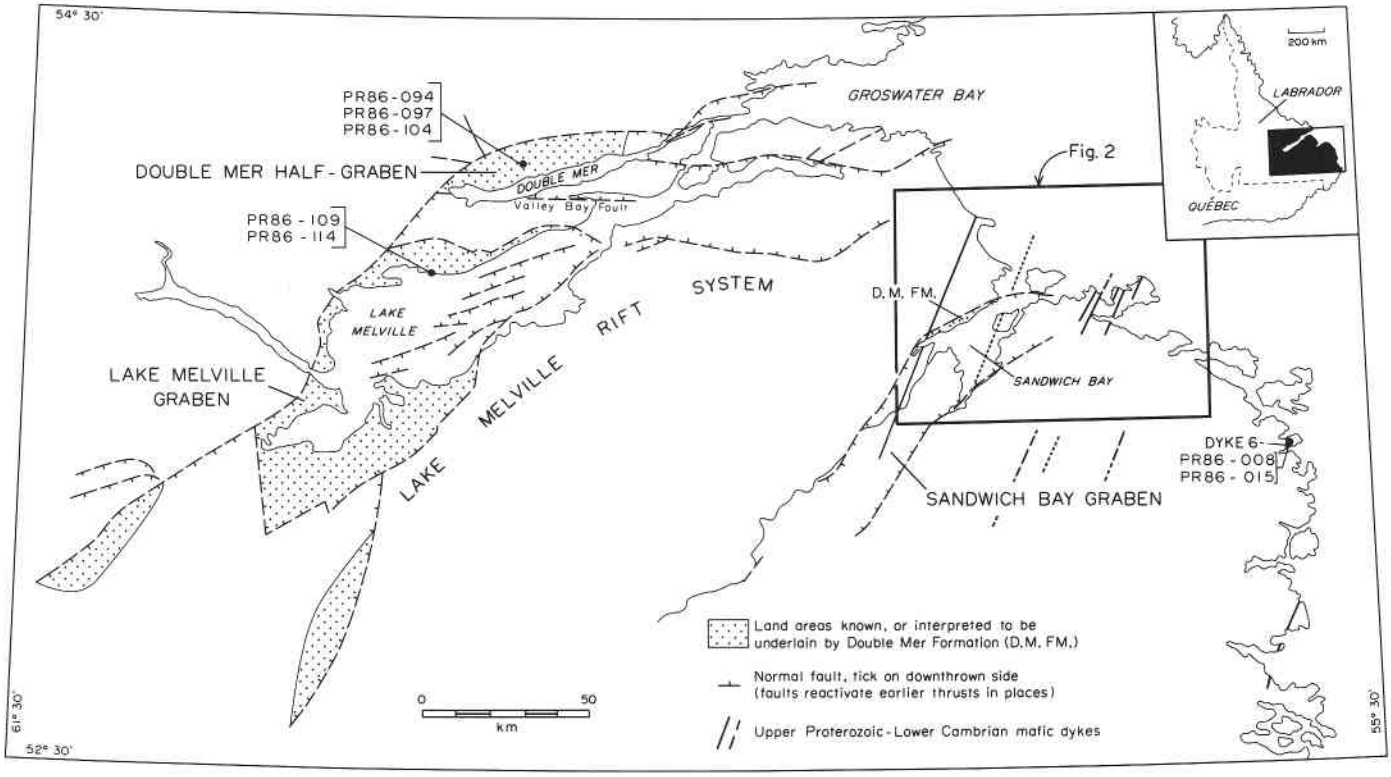


FIG. 1. Distribution of the Long Range dykes and the Double Mer Formation in eastern Labrador. Location of the Double Mer sites and two sites (PR86-008 and PR86-015) from dyke 6 are also shown.

U–Pb age obtained from Labrador is consistent with a 614 ± 10 Ma $^{40}Ar/^{39}Ar$ age obtained from a Long Range dyke in the Long Range Inlier of Newfoundland (Stukas and Reynolds 1974; recalculated using decay constants reported by Steiger and Jäger 1977). Previous K–Ar geochronological studies on Long Range dykes in Newfoundland yielded ages between 815 ± 35 Ma (Pringle *et al.* 1971) and 341 ± 100 Ma (Clifford 1965; Wanless *et al.* 1966), but the extreme ages were acknowledged to be improbably old or unrealistically young (cf. Bostock *et al.* 1983 for a summary of K–Ar geochronology for the Long Range dykes).

The Long Range dykes in southeast Labrador trend about 020°, having a small but distinct contrast in orientation to those of the Long Range Inlier, which trend at about 040°. This difference may simply indicate a regional fan pattern, especially as the most southeasterly dyke in Labrador has a trend intermediate between the more northerly dykes and those in the Long Range Inlier. Unlike the numerous, closely spaced dykes in the Long Range Inlier, the dykes in Labrador are not abundant and occupy 8–10 linear, partly en echelon, fracture systems. Inland, the dykes are only sporadically exposed, and their position must be inferred from aeromagnetic and topographic lineaments. On the basis of these criteria, the westernmost dyke has an interpreted length of about 120 km, and at least three other dykes have lengths exceeding 50 km. The dykes are near vertical and have widths varying from a few metres to over 100 m. They are typically coarse grained and are massive and have well-preserved ophitic textures. The degree of alteration is slight in some samples, whereas in others the olivine is almost totally altered to greenschist-facies assemblages. Dyke 6 (sites PR86-008 and PR86-015) is the most mineralogically modified, is foliated, and has been injected by quartz–epidote-filled veins.

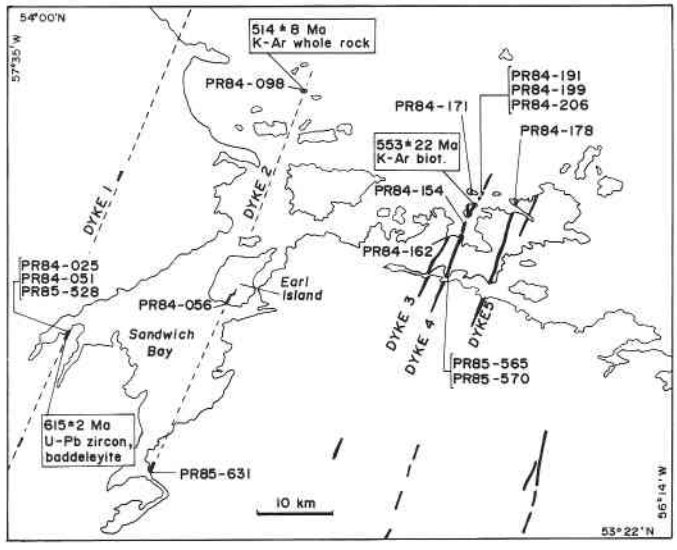


FIG. 2. Sketch map of the Sandwich Bay area in eastern Labrador, showing the sampling localities of most of the Long Range dykes. Radiometric dates, where available, are also indicated.

Double Mer Formation

The Double Mer Formation (Fig. 1) is an unmetamorphosed and undeformed sequence of post-Grenvillian rift-related red-bed arkosic sandstone, conglomerate, siltstone, and shale. The formation is closely associated with the Lake Melville rift sys-

tem, which extends inland for at least 300 km from the coast. It lacks fossils or radiometrically dateable material, and the only definite age constraint is that it postdates Grenvillian deformation. Gower *et al.* (1986) and Gower (1988) have discussed the distribution, characteristics, tectonic setting, and age of the Double Mer Formation. All red-bed samples for paleomagnetic study were collected from five sites within the Double Mer half graben and the northern part of the Lake Melville graben (Fig. 1), the only two areas having substantial thicknesses of exposed Double Mer Formation.

Paleomagnetism of the Long Range dykes

Analytical procedures

Ninety-nine oriented samples were collected from six mafic dykes at the sites shown in Figs. 1 and 2. Orientation of samples is by both solar and magnetic compasses, with good agreement between the two. Chilled margins were sampled wherever possible; when they were not exposed, our collection covered the full width of the dyke, including the coarser central parts. Where possible, various techniques (paleomagnetic, geochemical, geochronological) were applied to parts of the *same sample* to ensure that all data obtained were applicable to a specific rock.

A Schonstedt SSM-1A spinner magnetometer was used for remanence measurements. The specimens were subjected to both alternating-field (AF) and thermal (TH) demagnetizations using Schonstedt demagnetizers (models GSD-1 and TSD-1). Care was taken to keep the ambient field, during the demagnetization process, to less than 10 nT during AF treatment and to less than 5 nT during TH treatment. Each specimen was demagnetized in detail, and the specimen's response was individually analyzed to identify the various components of magnetization. The analytical procedures employed include visual inspection of the vector diagrams (Zijderveld 1975) for straight-line segments and their quantitative analysis by principal-component method (Kirschvink 1980), remagnetization-circle procedures of Halls (1976, 1978), and vector-subtraction procedures of Roy and Park (1974) and Hoffman and Day (1978). In using the principal-component analysis, the cutoff maximum angular deviation (MAD) value for accepting a linear segment on the vector plot is chosen as 10°. Rock magnetic studies were conducted to determine the nature of the magnetic minerals, and their domain state.

The Lowrie–Fuller test (Lowrie and Fuller 1971) of Fig. 3*a* shows that the isothermal remanence magnetization (IRM) is more resistive to AF demagnetization than the natural remanence magnetization (NRM), suggesting that remanence carriers are probably multidomain in nature, an inference consistent with the higher than normal values of susceptibilities for the samples and the fact that Koenigsberger ratios are less than 1. Samples got saturated IRM typically in fields of the order of 0.2 T (Fig. 3*b*). Back field of approximately 25 mT was required for complete removal of the IRM. Such behaviour is typical of samples carrying magnetite as the main magnetic mineral, an inference supported also from Curie temperature determinations (Fig. 3*c*), with a Curie point of 570°C noted for most of the samples.

Figure 4 shows the response of samples to demagnetization. Sample PR56A, subjected to AF treatment, exhibits an initial steep downward magnetization, followed by a characteristic remanence that is southeast and intermediate downward beyond 10 mT. The straight-line segment on the vector diagram passes through the origin, suggesting absence of other

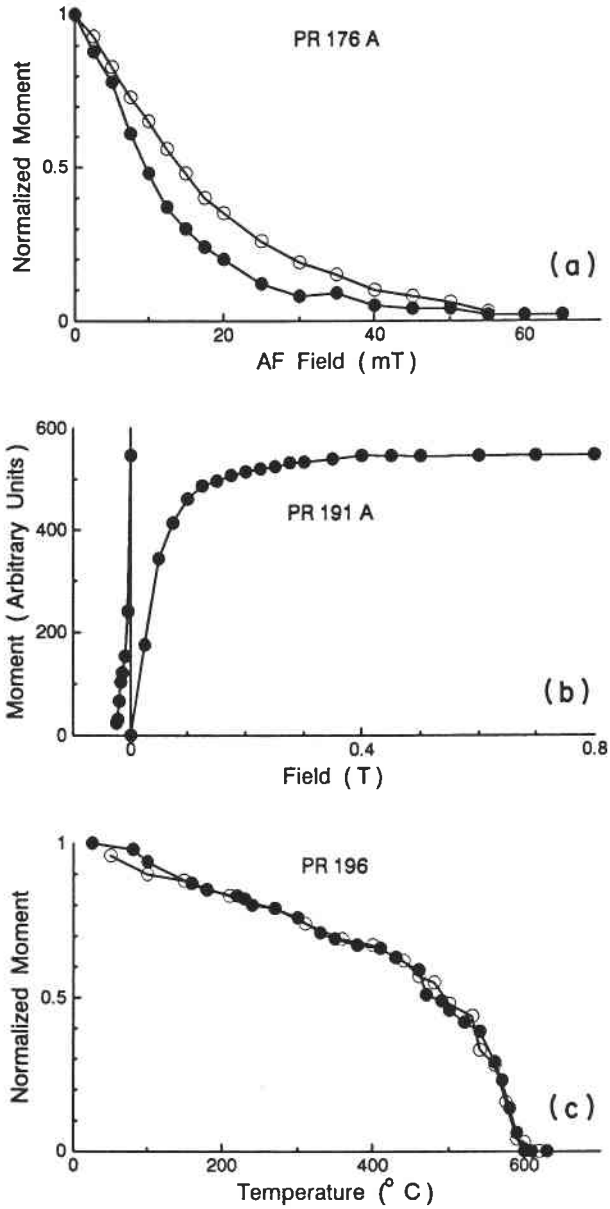


FIG. 3. (*a*) Comparison of response to AF demagnetization of NRM and IRM for specimen PR176A to infer the domain state of the remanence carriers. ●, IRM; ○, NRM. (*b*) IRM acquisition and back-field characteristics for specimen PR191A. (*c*) Magnetic moment vs. temperature (J_s-T curve) for sample PR196 in a field of 0.86 T. ●, heating; ○, cooling.

components. Samples PR64B and PR64C, subjected to TH and AF treatments, respectively, show similar behaviours, with east-southeasterly and downward magnetizations recognized in both. The remanent intensity dropped to less than measuring accuracy beyond 570°C, leaving no doubt that magnetite is the remanence carrier. Similar response of specimens to both kinds of demagnetization gives credibility to the recognized paleomagnetic directions. TH results often tended to be noisy, possibly because of acquisition of spurious components during the cooling process in the demagnetizer. The demagnetization treatment chosen for most samples was, hence, the AF procedure. Sample PR98A also showed a southeast and intermediate downward magnetization beyond 30 mT, after the initial removal of a steep downward compo-

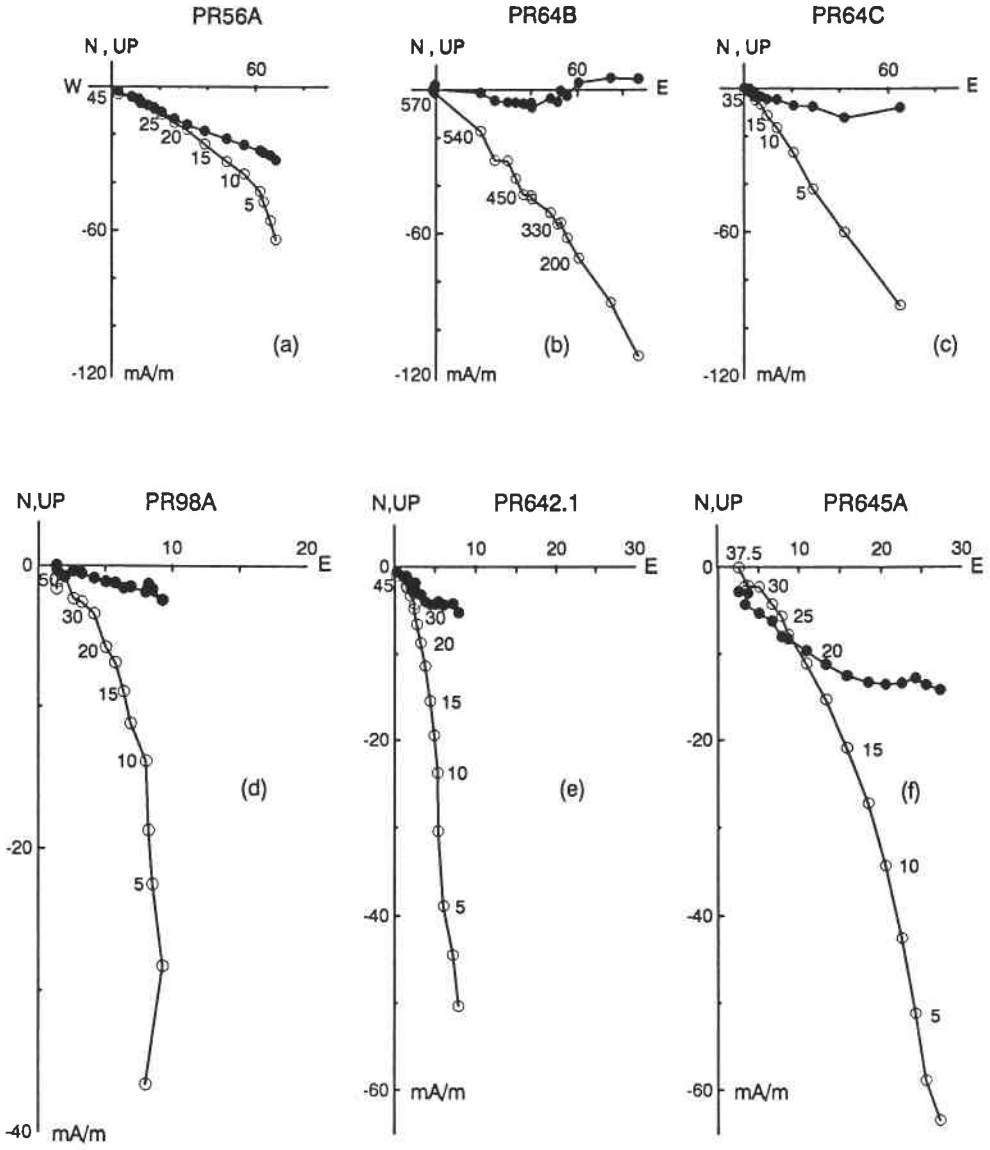


FIG. 4. Response of specimens from Long Range dykes to AF and TH demagnetizations in the form of vector diagrams. ●, projection in the horizontal plane; ○, projection in the east−west vertical plane. Numbers next to the open symbols indicate the field (mT) or temperature (°C).

nent. Specimens PR642.1 and PR645A show a similar trend with the characteristic remanence recognized at steps beyond 30 mT.

To test whether the observed magnetization is primary, baked contact tests were attempted wherever possible, and an example of this test is shown in Fig. 5. Sample locations from pre-Grenvillian gabbroic host rock and intruding dyke 2 are shown in Fig. 5*a*, with sample 108 next to the contact, 109 about 8.5 m from the dyke, and 107 at 13 m from the dyke. Specimen 108A (Fig. 5*c*) shows southeast and downward direction, similar to that of the dyke specimen (Fig. 5*b*). Magnetization of specimen 109A, from the hybrid zone, changed systematically from a steep downward direction to a northwest and upward sense during AF treatment (Fig. 5*d*). The high coercivity component has a northwest and upward direction that is the same as that characteristic for late Grenville times (Murthy *et al.* 1989*b*). Specimen 107A (Fig. 5*e*) exhibits typical Grenville remanence, stable, northwest, and upward. From the observed behaviour of the samples, it is concluded that the characteristic remanence in the dykes (shown in Table 1) is primary and representative of the paleomagnetic field at time of intrusion of the dykes.

Results

We have divided the paleomagnetic results (Table 1) for the Long Range dykes into two groups: (*i*) those that have a remanence direction in common (dykes 2, 4, and 6), and (*ii*) those that show "anomalous" remanences (dykes 1, 3, and 5). The mean characteristic remanence for the first group is $D = 124.8°$, $I = 55.5°$. We note that the K−Ar whole-rock age of 514 ± 8 Ma noted earlier was obtained from dyke 2, and the K−Ar biotite age of 553 ± 22 Ma comes from dyke 4. These dates are almost certainly postemplacement cooling ages and therefore also provide a minimum time for acquisition of the paleomagnetic signature. It should be noted that dyke 6, which has not been dated, shows some compositional differences with respect to dykes 2 and 4, particularly with respect to CaO, P_2O_5, F, Sr, Zr, and Ba (Table 3). Because

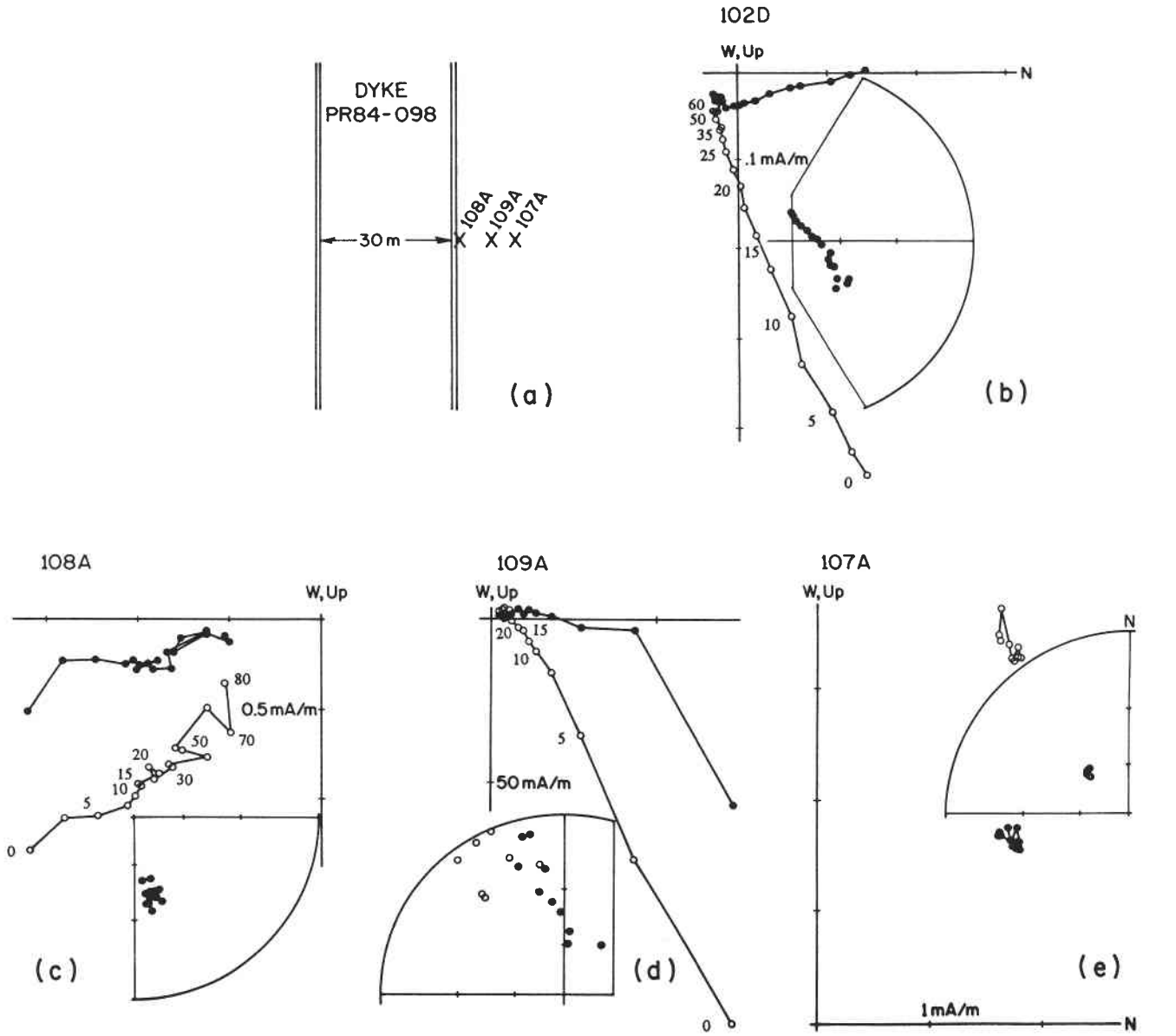


FIG. 5. Baked-contact test for dyke 2. (*a*) Dyke 2 to scale, with location of host-rock samples. (*b*) Response of dyke specimen 102D to demagnetization shown as a vector diagram and a stereographic projection using a Wulff's net. (*c*–*e*) Response of host-rock specimens 108A, 109A, and 107A to demagnetization presented as vector diagram and in stereographic projection. Vector diagram symbols as in Fig. 4. Stereographic projection: ●, positive inclinations; ○, negative inclinations.

the remanence is the same as that of dykes 2 and 4, and this dyke is the only one that is mineralogically altered, we suggest that the compositional modification is the result of low-grade metamorphism that the more westerly dykes escaped.

The anomalous remanences in the second group (dykes 1, 3, and 5) can be explained in two ways: either not all the dykes are of the same age, or the remanence directions represent disturbed data. On present data, we are obliged to reject the first alternative. Not only do the dykes have a common northeast trend, but they also have similar weathering appearance, mineral composition, and petrographic characteristics (grain size, texture, and mineral crystallization order). Also, their chemical composition is extremely similar to that of the first group, especially dykes 2 and 4 (Table 3). It is also unacceptable to dismiss the 615 ± 2 Ma age of dyke 1 as not representative of the Long Range swarm, given that the only other reliable date (an $^{40}Ar/^{39}Ar$ age of 614 ± 10 Ma), from a Long Range dyke in Newfoundland, is analytically identical.

If it is accepted that all the dykes belong to a single swarm, then reasons must be sought to explain the paleomagnetic behaviour of the "anomalous" dykes. The main phase of dyke 1 (sites PR84-025 and PR85-528) is coarse to very coarse grained. This was intruded by a fine-grained porphyritic phase (site PR84-051), which has the same trend. The coarse-grained phase was subsequently intruded by an east-southeast-trending Carboniferous dyke. Detailed study at this site showed the coarse-grained material close to the Carboniferous dyke had acquired a typical Carboniferous paleomagnetic signature, whereas more distant specimens yielded the directions quoted in Table 1. The remanence direction in dyke 1 cannot be dismissed as spurious, as a similar direction was obtained from dyke 3, and also in another dyke from Sandwich Bay (site

TABLE 1. Paleomagnetic data for Long Range dykes from southeast Labrador

Site	Samples	N	Field data: Trend of dyke (°)	Field data: Width of dyke (m)	Characteristic remanence: D (°)	I (°)	k	α_{95} (°)	n_1	n_2
DYKES CARRYING PRIMARY REMANENCE										
Dyke 2										
PR84-056	56–64	9	000	50	111.2	50.9	31	8.0	7	12
PR84-098	98–109	8	028	30	106.2	49.9	45	7.9	5	10
PR85-631	631–645	5	020	50	131.7	39.7	66	8.5	3	8
Mean (N = 3 sites)					117.3	47.4	52	17.3		
Dyke 4										
PR84-154	154–161	6	020	>50	114.2	52.4	60	9.5	3	6
PR84-171	171–177	5	000	35	158.0	53.2	32	22.3	3	0
PR84-191	191–198	6	045	50	129.0	48.6	44	18.8	3	0
PR84-199	199–205	5	017	0.25	120.5	51.7	16	15.6	4	8
PR84-206	206–213	5	020	>20	136.5	64.3	32	10.4	3	10
PR85-565	565–569	5	000	>1	137.1	40.0	104	6.6	4	4
PR85-570	570–576	5	025	>50	136.8	43.3	11	20.1	4	6
Mean (N = 7 sites)					133.1	51.2	48	8.8		
Dyke 6										
PR86-008	008–014	5	026	3	117.0	66.0	79	5.8	6	6
PR86-015	015–021	5	020	1	132.6	68.6	257	3.1	7	5
Mean (N = 2 sites)					124.6	67.5				
Mean characteristic remanence (N = 3 dykes)					124.8	55.5	48	18.0		
DYKES CARRYING ANOMALOUS REMANENCE										
Dyke 1										
PR84-025	025–030	5	025	>50	201.4	−72.1	103	7.2	0	3
PR85-528	528–534	4	025	>50	220.5	−85.6	6	28	0	4
PR84-051	051–055	8	025	0.3	225.5	−67.9	73	12	0	7
Mean (N = 3 sites)					215.3	−75.5	67	15.2		
Dyke 3										
PR84-162	162–170	6	035	>10	249.8	−49.2	43	16.4	2	4
Dyke 5										
PR84-178	178–184	5	023	60	279.1	28.1	31	7.3	8	15

NOTES: N, number of dyke samples collected and does not include the host-rock samples for the contact tests; D, declination; I, inclination; k, Fisher's precision parameter; α_{95}, radius of the 95% circle of confidence; n_1, number of specimens showing characteristic remanence by the stable end-point (vector plot segments leading to the origin) method; n_2, number of specimens yielding characteristic remanence by the great circle, vector subtraction, and other procedures.

PR84-083, D = 227°, I = −51°; Murthy *et al.* 1989*a*). Explanation of this "anomalous" behaviour awaits our further compilation of results from coastal Labrador.

Dyke 5 samples exhibited great stability. The direction obtained for this dyke (Table 1) cannot also be rejected as spurious, because similar westerly and shallow positive inclinations were detected earlier from Earl Island (location shown in Fig. 2) which Murthy *et al.* (1989*b*) suggested could be Elsonian. If the remanence in the Earl Island dykes and (the likely much younger) dyke 5 have a common cause, it cannot, of course, be linked to Elsonian events, but we do not yet have a satisfactory explanation for its acquisition.

Paleomagnetism of the Double Mer Formation

NRM intensities for the red-bed samples have values of approximately 2.0 mA/m. AF demagnetization did not produce significant decay of the natural remanence, which is to be expected if hematite is the remanence carrier. IRM and back-field experiments also showed this to be the case, with no saturation in magnetization achieved even at fields of 0.6 T and the requirement of more than 0.3 T back field to reverse the direction of original IRM. Figure 6 (*in situ* coordinates) shows results of thermal demagnetization of the Double Mer samples. A north- and downward-directed component was removed at temperatures up to 300°C. Beyond 580°C, the specimens exhibited southeast and downward directions, which when referred to the paleohorizontal (Table 2) are within error identical to the one carried by the dykes. The blocking temperatures are discrete, with substantial drop in intensity between 660 and 680°C. Although it cannot be said unambiguously that the recovered magnetization is primary, the grouping is tighter after referral to the bedding plane.

Discussion

The magnetization of the Double Mer Formation is probably ancient, since no late Paleozoic or Mesozoic directions are similar to the recovered one. A late Precambrian or Early Cambrian age has been generally assigned to the Double Mer Formation, mainly on the basis of comparison with the

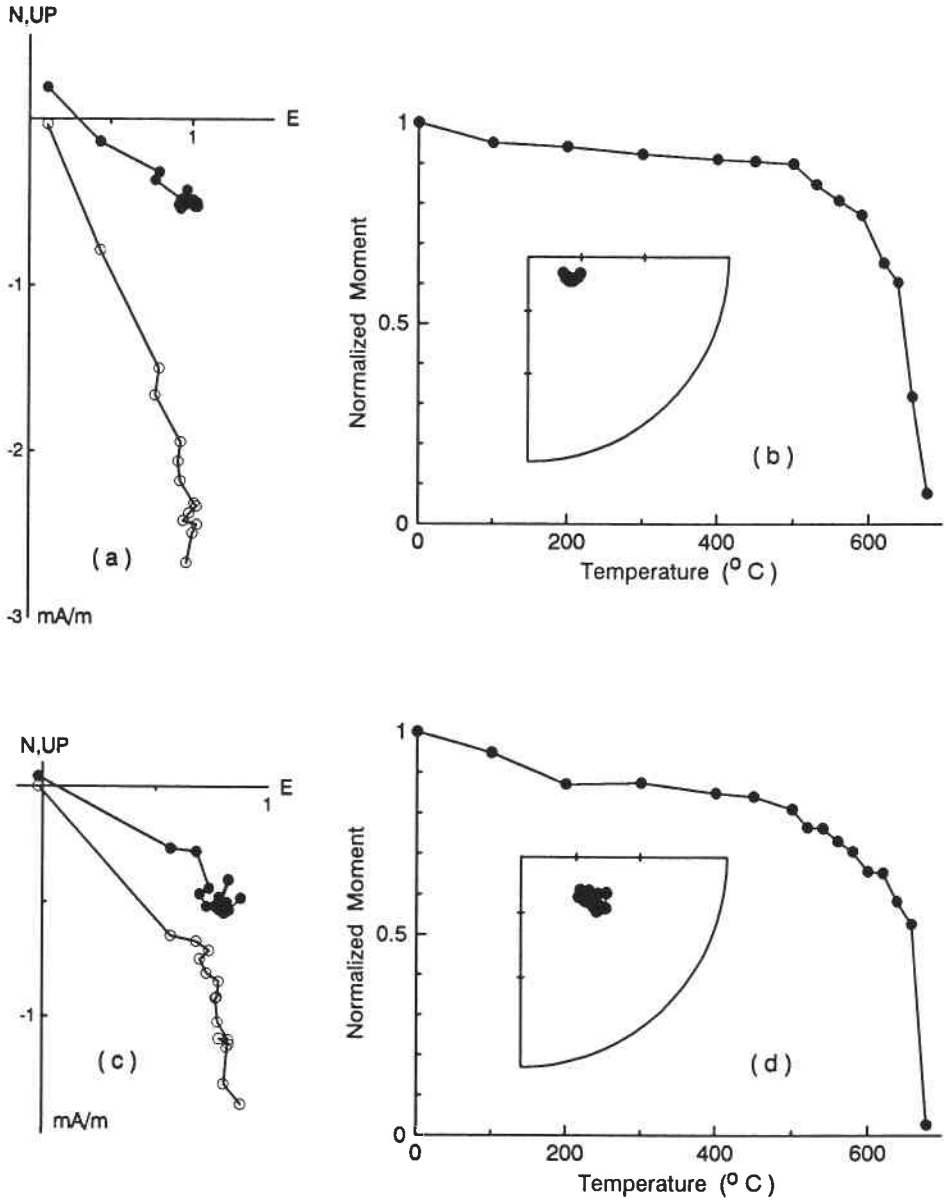


FIG. 6. Response of Double Mer sandstone specimens to thermal demagnetization up to 680°C. (*a*, *c*) Vector diagrams for 86-94.2 (*a*) and 86-102.1 (*c*). ●, projection in the horizontal plane; ○, projection in the east–west vertical plane. (*b*, *d*) Normalized intensity variation with temperature on demagnetization for 86-94.2 (*b*) and 86-102.1 (*d*). The insets show the stereonet representation of response of specimens to demagnetization. In each stereonet diagram, the last point of demagnetization at 680°C was not included for the sake of clarity of the diagram.

Bradore Formation (Labrador Group) along the Strait of Belle Isle. It is likely that both the Double Mer Formation and the Long Range dykes are related to the early stages of the rift–drift process (cf. Williams and Hiscott 1987) that led to the formation of the Iapetus Ocean. In the Strait of Belle Isle area, Grenvillian basement is overlain by terrestrial or shallow-marine conglomerate and arkose of the Bateau Formation. These rocks are intruded by Long Range dykes that acted as feeders for the overlying Lighthouse Cove basaltic flows, which in turn are overlain by red-bed arkose and conglomerate of the Bradore Formation. The minimum age of this sequence is late Early Cambrian (550 Ma), based on constraints imposed by paleontological data from limestones belonging to the overlying Forteau Formation.

Gower *et al.* (1986) proposed a structural model linking the configuration of the Lake Melville rift system with the trend of fractures now occupied by the Long Range dykes. No Long Range dykes have been found intruding the Double Mer Formation, but the fact that dykes are lacking cannot be taken as evidence that the Double Mer Formation is younger, as the Double Mer Formation and the mafic dykes do not occur in the same area. Combined with the dykes' magnetization and comparing it with the other Eocambrian data, it may be concluded that the Double Mer data represent an Eocambrian remanence and presentation of data with respect to ancient horizontal is justified.

The mean directions for three of the Long Range dykes and for the Double Mer sandstone relative to the bedding plane are identical (Fig. 7, inset). Paleopoles for the two rock units of this study, along with paleopoles of Eocambrian and Cambrian age from cratonic North America are plotted in Fig. 7. In each individual result quoted, the authors have argued for an

TABLE 2. Paleomagnetic data for the Double Mer Formation, southern Labrador

Site	Samples	N	Strike	Tilt	Characteristic remanence: Present horizontal D (°)	I (°)	k	α_{95} (°)	Tilt corrected D (°)	I (°)
PR86-094	094–098	5	317	17	134.9	64.8	13	19.7	103.7	59.4
PR86-099	099–103	5	317	17	142.3	56.8	9	20.5	117.2	54.6
PR86-104	104–108	5	317	17	129.8	65.3	21	11.3	99.7	58.6
PR86-109	109–113	5	268	9	111.0	19.3	5	27.0	107.8	22.6
PR86-114	114–118	5	268	9	132.3	41.3	9	29.2	123.7	50.3
Mean characteristic remanence										
Present horizontal	$N = 5$ sites; $D = 128.2°$; $I = 52.0°$; $k = 15.4$; $\alpha_{95} = 20.1°$									
Bedding plane	$N = 5$ sites; $D = 110.8°$; $I = 50.1°$; $k = 24.6$; $\alpha_{95} = 11.7°$									

NOTES: N, number of samples collected; D, declination; I, inclination; k, Fisher's precision parameter; α_{95}, radius of the 95% circle of confidence.

TABLE 3. Geochemical data for Long Range dykes from southeast Labrador

Dyke:	1	2	3	4	5	6
SiO_2 (wt.%)	49.13	46.95	48.95	48.05	46.90	49.88
TiO_2	2.67	2.62	1.90	2.35	3.19	2.67
Al_2O_3	17.25	17.11	14.67	17.55	17.18	14.52
Fe_2O_3	2.91	2.50	4.10	2.60	2.66	3.66
FeO	7.91	9.13	9.75	7.79	7.84	5.84
MnO	0.21	0.20	0.24	0.17	0.14	0.10
MgO	4.51	6.16	5.66	5.56	5.12	7.51
CaO	6.40	8.40	9.07	8.53	9.95	4.38
Na_2O	3.94	3.07	2.93	3.43	3.23	3.95
K_2O	1.88	1.45	1.26	1.32	1.06	1.45
P_2O_5	0.77	0.50	0.43	0.52	0.33	1.78
LOI	1.90	1.61	0.36	1.72	1.08	3.71
Total	99.47	99.68	99.32	99.59	98.68	99.43
Li (ppm)	20	15	9	18	12	39
F	878	463	517	514	298	2243
V	128	239	373	183	264	185
Cr	76	57	104	58	125	183
Ni	45	87	53	74	60	92
Cu	28	57	136	39	63	29
Zn	163	106	131	108	193	142
Ga	33	29	20	29	30	28
Rb	39	33	17	28	19	34
Sr	570	623	524	680	717	936
Y	29	21	32	22	17	25
Zr	318	149	127	153	131	462
Nb	26	16	nd	17	13	15
Mo	5	4	4	4	4	4
Ba	531	426	873	499	403	1627
La	36	21	20	24	17	79
Ce	82	53	48	62	44	171
Pb	nd	nd	nd	nd	nd	nd
Th	nd	nd	nd	nd	nd	nd
U	0.9	0.4	0.5	0.5	0.2	1.2
N	2	3	1	7	1	2

NOTES: LOI, loss on ignition; N, number of analyses; nd, not detected.

Eocambrian–Cambrian age for the magnetization. Placement of the Double Mer paleopole in the middle of this grouping provides quantitative justification that this formation was deposited during Eocambrian times. Perusal of discussions on ages of magnetizations presented in Fig. 7 by original authors reveals *generally* that the paleopoles in the equatorial cluster are older than the poles farther to northwest. The northwestern grouping is close to late Paleozoic remagnetizations described in the literature, and some of the paleopoles may likely be contaminated by that effect.

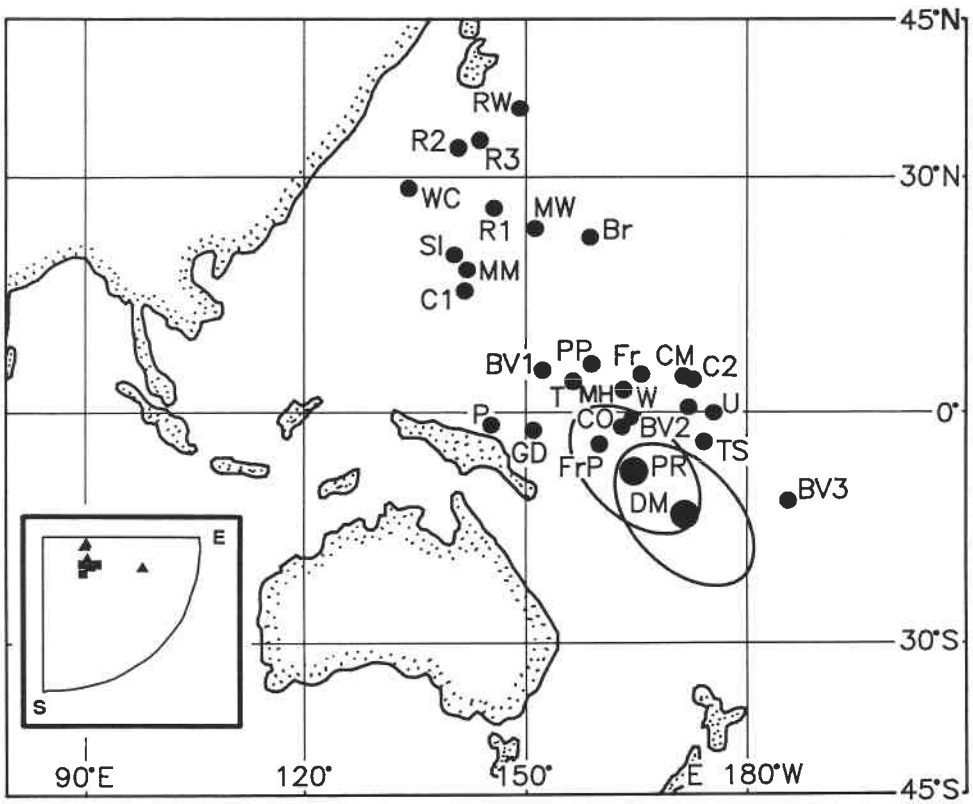


FIG. 7. Eocambrian and Cambrian paleopoles relative to cratonic North America. Ovals of confidence around the two poles obtained in the present study are also shown. The pole nomenclature is as follows: Br, Bradore Formation, Newfoundland (Deutsch and Rao 1977*a*); BV1, BV2, and BV3, Buckingham volcanics I, II, and III (Dankers and Lapointe 1981); C1 and C2, Colorado intrusives (French *et al.* 1977); CM, Cloud Mountain basalt (Deutsch and Rao 1977*b*); CO, Coronation sills (Robertson ad Baragar 1972); DM, Double Mer sandstone (this study); Fr, Franklin diabase (Fahrig and Schwarz 1973); FrP, Franklin lavas and diabases (Palmer and Hayatsu 1975); GD, Grenville dykes (Murthy 1971); MH, Moore Hollow Group, Texas (Farr and Gose 1991); MM, McClure Mountain Complex (Lynnes and Van der Voo 1984); MW, Morgan Creek – Welge, Texas (Watts *et al.* 1980*b*); P, Port au Port peninsula rocks (Prasad 1986); PP, Point Peak Member, Texas (Van der Voo *et al.* 1976); PR, Paradise River Long Range dykes (this study); R1, R2, and R3, Riley Formation, Texas (Watts *et al.* 1980*b*); RW, Rome–Waynesboro formations (Watts *et al.* 1980*a*); SI, Sept Isle intrusion (Tanczyk *et al.* 1987); T, Tapeats sandstone (Elston and Bressler 1977); TS, Taum Sauk limestone (Dunn and Elmore 1985); U, Unicoi basalt (Brown and Van der Voo 1982); W, Wichita granites (Spall 1970); WC, Wood Canyon Formation (Gillett and Van Alstine 1979). (Inset) Characteristic remanences isolated for the three "regular" Long Range dykes (■) and for the Double Mer sandstone sites ▲ plotted on a Wulff's equal-angle projection. All are positive inclinations.

Paleolatitude is a fundamental parameter in the discussion of tectonic relationships between cratonic landmasses. The paleolatitude of eastern North America, based on results of the present study and also the paleopoles included in the southern cluster of Fig. 7, was estimated as 26 ± 5°S, a figure consistent with earlier estimates (e.g., Irving 1979; Johnson and Van der Voo 1986). Comparing this with our earlier estimates for the Carboniferous (Murthy *et al.* 1989*a*), there is a net 10° northward movement of the craton from Eocambrian to Carboniferous times. The track of this movement can be estimated by investigating rocks of mid-Paleozoic age from the craton.

Results for Long Range dykes from the type locality in the Long Range Inlier of western Newfoundland are relevant in discussion of our dyke results. Preliminary results for the Long Range dykes in Newfoundland obtained earlier (Pullaiah *et al.* 1979) were, however, inconclusive. The steep, downward-directed magnetization isolated for the Newfoundland dykes need to be reevaluated using the more sophisticated techniques that are now available.

Conclusions

Three of six Long Range dykes in Labrador show a mean remanence direction of declination of (D) = 125°, inclination (I) = 56°. The time of acquisition of this direction predated 553 Ma, based on a postemplacement K–Ar biotite age obtained earlier from one of the investigated dykes. "Anomalous" remanence directions were obtained from three other Long Range dykes, including one whose emplacement is dated at 615 Ma (zircon and baddeleyite). The significance of the anomalous directions is uncertain, but they cannot be dismissed as spurious because they have also been found in other rocks in the region.

The Double Mer Formation shows a mean remanence direction of D = 111°, I = 50°. The similarity of this result to that obtained from the "regular" Long Range dykes is confirmation that both directions are Eocambrian. Using these data, the Eocambrian paleolatitude of eastern North America is calculated to be 26 ± 5°S.

A steep downard-directed magnetization isolated for the

Newfoundland dykes was obtained prior to the advent of the presently used analytical techniques in paleomagnetism, and when the laboratory procedures were much less sophisticated. A further paleomagnetic study of the Newfoundland Long Range dykes is necessary, as such comparisons would help in correlations and recognition of these very widely distributed dykes as one single swarm.

Acknowledgments

Mr. Tom Vallis at Memorial University of Newfoundland prepared the samples for laboratory measurements. The authors are grateful to the Journal referees, Ken Buchan and Henry Halls, for constructive suggestions which resulted in improvement of the paper. This paper is published with the permission of the Assistant Deputy Minister, Geological Surveys Branch, Newfoundland Department of Mines and Energy. Natural Sciences and Engineering Research Council of Canada support to GM prior to 1988 was helpful in the fieldwork that formed the basis of the results reported here. GM also wishes to acknowledge Lithoprobe and Energy, Mines and Resources Canada for support during fiscal 1990–1991 which helped in laboratory processing of the data and final preparation of the manuscript.

Bostock, H. H., Cumming, L. M., Williams, H., and Smyth, W. R. 1983. Geology of the Strait of Belle Isle, northwestern insular Newfoundland, southern Labrador, and adjacent Quebec. Geological Survey of Canada, Memoir 400.

Brown, P. M., and Van der Voo, R. 1982. Paleomagnetism of the latest Precambrian/Cambrian Unicoi basalts from the Blue Ridge, northeast Tennessee and southwest Virginia: evidence for Taconic deformation. Earth and Planetary Science Letters, **60**: 407–414.

Clifford, P. M. 1965. Paleozoic flood basalts in northern Newfoundland and Labrador. Canadian Journal of Earth Sciences, **2**: 183–187.

Dankers, P., and Lapointe, P. 1981. Paleomagnetism of Lower Cambrian volcanics and a cross-cutting Cambro-Ordovician diabase dyke from Buckingham (Quebec). Canadian Journal of Earth Sciences, **18**: 1174–1186.

Deutsch, E. R., and Rao, K. V. 1977*a*. Paleomagnetism of the Lower Bradore sandstones, and the rotation of Newfoundland. Tectonophysics, **33**: 337–357.

Deutsch, E. R., and Rao, K. V. 1977*b*. New paleomagnetic evidence fails to support rotation of western Newfoundland. Nature (London), **266**: 314–318.

Dunn, W. J., and Elmore, R. D. 1985. Paleomagnetic and petrographic investigation of the Taum Sauk limestone, southeast Missouri. Journal of Geophysical Research, **90**: 11 469 – 11 483.

Elston, D. P., and Bressler, S. L. 1977. Paleomagnetic poles and polarity zonation from Cambrian and Devonian strata of Arizona. Earth and Planetary Science Letters, **36**: 423–433.

Fahrig, W. F., and Schwarz, E. J. 1973. Additional paleomagnetic data on the Baffin diabase dykes and a revised Franklin pole. Canadian Journal of Earth Sciences, **10**: 576–581.

Fahrig, W. F., and West, T. D. 1986. Diabase dyke swarms of the Canadian Shield. Geological Survey of Canada, Map 1627A.

Farr, M. R., and Gose, W. A. 1991. Paleomagnetism of the Cambrian Moore Hollow Group, Texas: evidence for a primary magnetization carried by detrital magnetite. Journal of Geophysical Research, **96**: 9895–9907.

French, R. B., Alexander, D. H., and Van der Voo, R. 1977. Paleomagnetism of Upper Precambrian to Lower Paleozoic intrusive rocks from Colorado. Geological Society of America Bulletin, **88**: 1785–1792.

Gillett, S. L., and Van Alstine, D. R. 1979. Paleomagnetism of Lower and Middle Cambrian sedimentary rocks from the Desert Range, Nevada. Journal of Geophysical Research, **84**: 4475–4489.

Gower, C. F. 1988. The Double Mer Formation. *In* Late Proterozoic stratigraphy of the northern Atlantic regions. *Edited by* J. A. Winchester. Blackie and Son Ltd., London, pp. 113–118.

Gower, C. F., Erdmer, P., and Wardle, R. J. 1986. The Double Mer Formation and the Lake Melville rift system, eastern Labrador. Canadian Journal of Earth Sciences, **23**: 359–368.

Grasty, R. L., Rucklidge, J. C., and Elders, W. A. 1969. New K–Ar age determinations on rocks from the east coast of Labrador. Canadian Journal of Earth Sciences, **6**: 340–344.

Halls, H. C. 1976. A least-squares method to find a remanence direction from converging remagnetization circles. Geophysical Journal of the Royal Astronomical Society, **45**: 297–304.

Halls, H. C. 1978. The use of converging remagnetization circles in paleomagnetism. Physics of the Earth and Planetary Interiors, **16**: 1–11.

Hoffman, K. A., and Day, R. 1978. Separation of multi-component NRM: a general method. Earth and Planetary Science Letters, **40**: 433–438.

Irving, E. 1979. Paleopoles and paleolatitudes of North America and speculations about displaced terrains. Canadian Journal of Earth Sciences, **16**: 669–694.

Johnson, R. J. E., and Van der Voo, R. 1986. Paleomagnetism of the Late Precambrian Fourchu Group, Cape Breton Island, Nova Scotia. Canadian Journal of Earth Sciences, **23**: 1673–1685.

Kamo, S., Gower, C. F., and Krogh, T. E. 1989. A birthdate for the Iapetus ocean? A precise U–Pb zircon and baddeleyite age for the Long Range dykes, S.E. Labrador. Geology, **17**: 602–605.

Kirschvink, J. L. 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophysical Journal of the Royal Astronomical Society, **62**: 699–718.

Lowrie, W., and Fuller, M. 1971. On the alternating field demagnetization characteristics of multidomain thermoremanent magnetization in magnetite. Journal of Geophysical Research, **76**: 6339–6349.

Lynnes, C. S., and Van der Voo, R. 1984. Paleomagnetism of the Cambro-Ordovician McClure Mountain alkalic complex, Colorado. Earth and Planetary Science Letters, **71**: 163–172.

Murthy, G. S. 1971. The paleomagnetism of diabase dykes from the Grenville Province. Canadian Journal of Earth Sciences, **8**: 802–812.

Murthy, G., Gower, C., Tubrett, M., and Pätzold, R. 1989*a*. Paleomagnetism and geochemistry of Carboniferous Sandwich Bay dykes from coastal Labrador. Canadian Journal of Earth Sciences, **26**: 2278–2291.

Murthy, G., Gower, C., Tubrett, M., and Pätzold, R. 1989*b*. Paleomagnetism of pre-Grenvillian mafic intrusions from the Grenville Province, southeast Labrador. Canadian Journal of Earth Sciences, **26**: 2541–2555.

Palmer, H. C., and Hayatsu, A. 1975. Paleomagnetism and K–Ar dating of some Franklin lavas and diabases, Victoria Island. Canadian Journal of Earth Sciences, **12**: 1439–1447.

Prasad, J. N. 1986. A paleomagnetic investigation of Early Paleozoic rocks in western Newfoundland. Ph.D. thesis, Department of Earth Sciences, Memorial University of Newfoundland, St. John's, Nfld.

Pringle, I. R., Miller, J. A., and Warrell, D. M. 1971. Radiometric age determinations from the Long Range Mountains, Newfoundland. Canadian Journal of Earth Sciences, **8**: 1325–1330.

Pullaiah, G., Rao, K. V., Deutsch, E. R., *et al.* 1979. Paleomagnetism of the late Precambrian Long Range dykes, Newfoundland: an Acadian remagnetization? Abstracts, Canadian Geophysical Union, Annual Meeting, Fredericton, N.B.

Robertson, W. A., and Baragar, W. R. A. 1972. The petrology and

paleomagnetism of the Coronation sills. Canadian Journal of Earth Sciences, **9**: 123–140.

Roy, J. L., and Park, J. K. 1974. The magnetization process of certain redbeds: vector analysis of chemical and thermal results. Canadian Journal of Earth Sciences, **11**: 437–471.

Spall, H. 1970. Paleomagnetism of basement granites in southern Oklahoma: final report. Oklahoma Geology Notes, **30**: 136–150.

Steiger, R. H., and Jäger, E. 1977. Subcommission on Geochronology: convention on the use of decay constants in geo- and cosmochronology. Earth and Planetary Science Letters, **36**: 359–362.

Stukas, V., and Reynolds, P. H. 1974. $^{40}Ar/^{39}Ar$ dating of the Long Range dikes, Newfoundland. Earth and Planetary Science Letters, **22**: 256–266.

Tanczyk, E. I., Lapointe, P., Morris, W. A., and Schmidt, P. W. 1987. A paleomagnetic study of the layered mafic intrusion at Sept-Isles, Quebec. Canadian Journal of Earth Sciences, **24**: 1431–1438.

Van der Voo, R., French, R. B., and Williams, D. W. 1976. Paleomagnetism of the Wilberns Formation (Texas) and the Late Cambrian paleomagnetic field for North America. Journal of Geophysical Research, **81**: 5633–5638.

Wanless, R. K., Stevens, R. D., Lachance, G. R., and Rimsaite, J. Y. H. 1966. Age determinations and geological studies: K–Ar isotope ages. Geological Survey of Canada, Paper 65-17.

Wanless, R. K., Stevens, R. D., Lachance, G. R., and Delabio, R. N. 1970. Age determinations and geological studies. Geological Survey of Canada, Paper 69-2A.

Watts, D. R., Van der Voo, R., and French, R. B. 1980*a*. Paleomagnetic investigation of the Cambrian Waynesboro and Rome Formations of the Valley and Ridge Province of the Appalachian Mountains. Journal of Geophysical Research, **85**: 5331–5343.

Watts, D. R., Van der Voo, R., and Reeve, S. C. 1980*b*. Cambrian paleomagnetism of the Llano uplift, Texas. Journal of Geophysical Research, **85**: 5316–5330.

Williams, H., and Hiscott, R. N. 1987. Definition of the Iapetus rift–drift transition in western Newfoundland. Geology, **15**: 1044–1047.

Zijderveld, J. D. A. 1975. Paleomagnetism of the Esterel rocks, Ph.D. thesis, Department of Geology, University of Utrecht, Utrecht, The Netherlands.